Scheme and answer key for QP Code: **25900556**
MGU-UGP (Honours) Regular Examination OCTOBER 2025
Third Semester: Discipline Specified Course: MG3DSCPHY202-ATOMIC AND MOLECULAR SPECTROSCOPY
Duration 1 Hour 30 Minutes Maximum Marks: 50
Part A: Answer Any **10** Questions. Each question carries **2** Marks (2 x10 = 20)

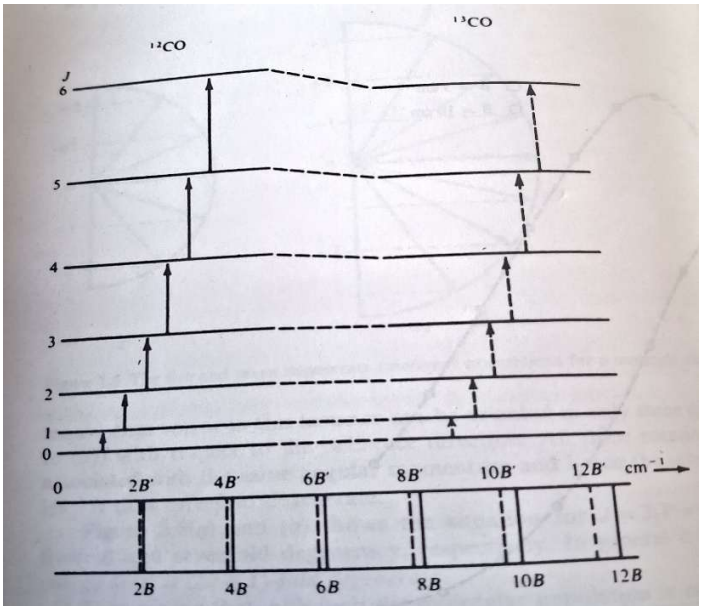
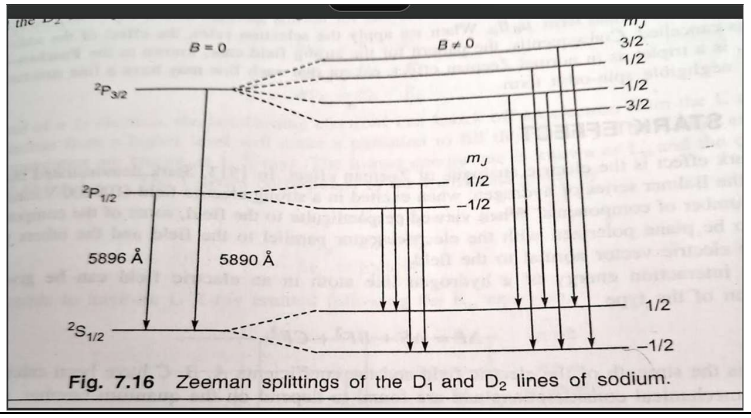
1	Term symbols are used to describe the quantum state of an atom. They give the information about total spin, total orbital angular momentum and total angular momentum of the atom. OR $^{2S+1}L_J$ S -Total spin quantum number L -Total orbital angular momentum quantum number J -Total angular momentum quantum number. 2S+1- Spin multiplicity	Statement with explanation- 2 marks Symbolic representation 1 Mark Terms explanation 1 Mark
2	Translational Energy (E_t) - Due to the motion of the centre of mass of the molecule. Vibrational Energy (E_v) - Due to the vibrational motion of atoms in the molecule. Electronic Energy (E_e) - Due to the motion of electrons in the molecule. Rotational Energy (E_r) - Due to the rotational motion of molecules about their centre of mass. E total = Et+ Ee + Ev+ Er	Each ½ Mark If represented by equation only- 1 Mark
3	A nucleus is NMR active if it has a non-zero nuclear spin I ≠0, which gives it a magnetic moment. Nuclei with odd numbers of protons or neutrons are NMR active.	Any Proper explanation 2 Marks. Definition of NMR- 1 mark
4	Expression for energy, $E = E_o+\mu_B B_o (m_l +2m_s)$ $m_s = \pm ½$ $E = E_o + \mu_B B_o (m_l \pm 1) = E = E_o + \mu_B B_o m_l \pm \mu_B B_o$	Equation only 1 Mark Substitution and final answer 2 Marks
5	It explains only hydrogen-like atoms and fails for multi-electron atoms. It does not explain the fine structure and Zeeman effect or Stark effect. It does not obey the uncertainty principle of Heisenberg. It cannot explain the relative intensities of spectral lines.	Any two limitations 2 Marks

6	jj coupling is a type of atomic coupling scheme used to describe the interaction between angular momenta in many-electron atoms, especially heavy atoms, where the spin-orbit interaction is strong. $J_1=L_I+S_1$; $J_2=L_2+S_2$; $J_3=L_3+S_3$. $J= \Sigma Ji$	Equation only 1 Mark, with Proper explanation 2 Mark
7	When all three principal moments of inertia of a molecule are equal, it is called a spherical top. $I_a = I_b = I_c$, therefore CH_4 is considered as a spherical top molecule.	Equation with explanation or statement 2 Marks
8	NMR - the radiofrequency region. ESR- the microwave region.	Each- 1 mark,
9	$\mu L = [\ell (\ell+ 1)]^{1/2}\mu_B$ $\ell=1$ $\mu L = [1 (1+ 1)]^{1/2}\mu_B = [2]^{1/2}\mu_B$	Equation only 1 mark Equation with proper substitution and correct result -2 marks
10	Rule of mutual exclusion- 'If a molecule has a centre of symmetry, Raman active vibrations are infrared inactive and vice versa.	Correct Statement – 2 Marks
11	The interaction between orbital angular momentum 'L' and spin angular momentum 'S' is called the spin-orbit interaction.	Correct Statement – 2 Marks
12	Phosphorescence is the delayed emission of light Emitted energy is less than the absorbed energy. Substance continues to glow even after the light source is removed.	A suitable definition containing two key points – 2 Marks
13	Electron Spin Resonance (ESR) is a technique confined to the study of molecules or ions having one or more unpaired electrons. Stable and unstable free radicals, transition metal ions, rare earth ions etc., are some of the systems investigated using ESR.	Any relevant explanation – 2 Marks
14	Scattering without a change in frequency is known as Rayleigh Scattering. The lines having frequencies on the low and high frequency side of the exciting line are called Stokes and anti-Stokes lines. These weak lines of modified frequencies are referred to as the Raman lines or Raman spectrum and the frequency shift from the exciting line is called Raman Shift.	Any relevant explanation – 2 Marks

Part B: Answer Any **6** Questions. Each question carries **5** Marks (5 x 6 = 30)

15.	**Normal Zeeman Effect**: The splitting of the energy levels or spectral lines of a substance placed in a magnetic field is called the Zeeman effect. In certain cases, the line is split into three lines called the normal Zeeman effect. Caused only by orbital angular momentum of the electron, the Spin of the electron is not considered. Interaction between the magnetic field and orbital angular momentum only. Spectral line splits into three components- one (πcomponent) two (σ components) Occurs when S = 0 Eg: $6^1D_2 \rightarrow 5^1P_1$ line of cadmium (Any similar example) **Anomalous Zeeman Effect:** The phenomenon of splitting of spectral lines of atoms having non-zero angular momentum in the presence of a magnetic field is often more complex than the three lines observed in the normal Zeeman effect. It is caused by both orbital and spin magnetic moments and it results in the interaction between the magnetic field and total angular momentum (J = L + S) Spectral lines splits into more than 3 components Occurs when S≠0 Eg; $^2p_{3/2} \rightarrow ^2s_{1/2}$ line of sodium (Any similar example)	Explanation with any one example – 2.5 Marks each
16.	Speed of light c= $3x10^8$ m/s Wavelength, λ= 460nm=460 $x10^{-9}$ m Frequencyν = c/ λ = $3x10^8/460\ x10^{-9}$ = 6.52 $x10^{14}$ Hz Energy = $h\nu$ Planck's constant, h = 6.62607015 × 10^{-34} Js E = 6.62607015 × 10^{-34}×6.52 $x10^{14}$ = 4.32×10^{-19} J OR 2.70 eV 460nm lies in the blue region of the visible spectrum.	Equations -2 marks Substitution and final answer with unit 2 Marks Colour- 1 Mark

17	Zero-point energy is the minimum energy possessed by a quantum mechanical system even at absolute zero temperature (0 K). The energy levels of a harmonic oscillator is given by $E_n = (n + ½)\ h\nu$ where n = 0, 1, 2, 3… h = Planck's constant ν-frequency for ground state, n =0 $E_o = (0 + ½)\ h\nu = ½\ h\nu$ This Energy $E_o = ½\ h\nu$ is called zero-point energy. No system can have zero energy - even at 0 K, a particle retains its energy due to quantum fluctuations.	Statement 1.5 Marks Derivation of Expression 2 Marks Significance 1.5 Marks
18	An important property of a nucleus is its spin angular momentum $I = [i(i+1)]^{½}\ ħ$, or simply iħ where i is the spin quantum number, which can have half-integral or integral values, including zero. Magnetic moment, $\mu = g_N\mu_N\ I/\ ħ = g_N\mu_N i$ $\mu N = eħ/2m_p = 5.0508 \times 10^{-27}$ J/T Consider a nucleus having a magnetic moment μ that is placed in a magnetic field B_0. The interaction energy is given by, $E = -\mu.B_0$ Substituting, we get $E_{-1/2} = ½\ g_N\ \mu_N\ B_0$ and $E_{+1/2} = -½\ g_N\ \mu_N\ B_0$ $h\nu = g_N\ \mu_N\ B_0 = 2\mu B_0$	Statement only 2.5 Marks Statement and derivation -5 Marks
19	i) Lyman Series: - All transitions which end at n_f=1 from excited states n_i= 1, 2 ,3 …∞; Comes under ultra violet spectral region, equation (ii) Balmer series: - Consists of wave numbers emitted when electron transitions end at n_f=2 from excited state n_i= 3,4,5---; Comes under visible spectral region iii) Paschen Series: Comes under Infra-red spectral region, n_i = 4,5,6… iv) Brackett Series: Comes under Infrared spectral region , n_i = 5,6,7… v) Pfund Series: - Comes under Infrared spectral region n_i= 6,7,8…	Name and any one correct point on each series- 1 mark each. Only names – ½ mark each

20	 Sample diagram is given	Spectra of rigid diatomic molecule- 3 marks. Spectra due to isotope- 2 marks
21	 **Fig. 7.16** Zeeman splittings of the D_1 and D_2 lines of sodium.	Figure with explanation 5 Marks
22.	Transitions between rotational levels within a given vibrational level give the pure rotational spectra in the far infrared or microwave region. Transition between vibrational levels within the same electronic state constitutes the vibrational spectra. This usually falls in the infrared region. Transitions between electronic energy levels give the spectrum in the visible or ultraviolet region, which is known as the electronic spectrum. Explanation using a diagram can also be considered.	Explanation for three spectra 5 Marks